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COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./ DUE DATE
NSF 23-603 10/21/2026

☐ Special Exception to Deadline Date Policy

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NSF PROPOSAL NUMBER

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NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE
Varvak, Edward A

AWARDEE ORGANIZATION CODE (IF KNOWN)
P270144017

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Chicago,IL 60637 US

NAME OF PRIMARY PLACE OF PERF
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TITLE OF PROPOSED PROJECT
Postdoctoral Fellowship: MSPRF: The Hodge theory of Lagrangian fibrations, with applications to hyper-Kahler geometry

SHOW LETTER OF INTENT ID IF APPLICABLE

REQUESTED AMOUNT	PROPOSED DURATION (1-60 MONTHS)	REQUESTED STARTING DATE	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE
$ **190,000**	48 months	**09/06/2027**	

THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW

☒ TYPE OF PROPOSAL **Postdoctoral Fellowship**
☒ COLLABORATIVE STATUS **Non-Collaborative**
☐ BEGINNING INVESTIGATOR
☐ DISCLOSURE OF LOBBYING ACTIVITIES
☐ PROPRIETARY & PRIVILEGED INFORMATION
☐ HISTORIC PLACES
☐ LIVE VERTEBRATE ANIMALS IACUC App. Date _____
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☐ POTENTIAL LIFE SCIENCES DUAL USE RESEARCH OF CONCERN
☐ OFF-CAMPUS OR OFF-SITE RESEARCH
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PI/PD DEPARTMENT

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Application Form for Mathematical Sciences Postdoctoral Research Fellowships (MSPRF) and Postdoctoral Research Fellowships in Biology (PRFB)

Questions for MSPRF and PRFB

Doctoral Degree

Organization Name: **Univerity of Chicago**

Department Name: **Mathematics**

Date Awarded or Anticipated Date: **06/05/2027**

Current Position

Organization Name: **University of Chicago**

Organization Type: **Private (PRIV)**

Department Name: **Mathematics**

Position Title: **Graduate Student**

Position Start Date: **09/28/2026**

Other Support

Applied for any other fellowships or similar appointments for all or part of your fellowship duration: **No**

Agency or Program Name:

Previously applied for or received a NSF postdoctoral fellowship: **No**

Date Applied or Awarded:

Program Name:

Sponsoring Scientist(s)

Name:

Organization:

Name:

Organization:

Name:

Organization:

Name:

Organization:

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Additional Questions for MSPRF Only

Research Classification

Primary Classification: **14 - Algebraic geometry**

Secondary Classification: **11 - Number theory**

Secondary Classification:

Secondary Classification:

Area of Proposed Research

Area of Mathematical Sciences: **Pure Mathematics**

Plan of Fellowship Use

Program Options and Duration: **The Research Fellowship for 18 academic months full-time**

Have you secured a suitable commitment for a teaching position at your proposed place of performance:

Short Term International Experience

Short Term International Experience Proposed: **No**

Organization Name:

Sponsor:

Justification

Your proposed place of performance should be different from where you received your doctoral degree or where you are currently employed, otherwise a justification may be needed.

My proposed place of performance is different from where I received my doctoral degree and different from where I am currently employed

Justification:

Overview

The PI will study the Hodge theory of Lagrangian fibrations at MIT, with sponsoring scientist Davesh Maulik. Hyper-Kähler manifolds are basic building blocks of compact Kähler manifolds with trivial canonical bundle, and classifying them is a fundamental open problem. By the recently proven hyper-Kähler SYZ conjecture, every known deformation type of hyper-Kähler manifolds admits Lagrangian fibrations, whose general fibers are abelian varieties of half the dimension. The project asks which abelian varieties occur as such fibers, i.e. where the associated period map lands.

The PI's thesis proves that for a non-isotrivial Lagrangian fibration of a $2n$-dimensional hyper-Kähler manifold, the monodromy is irreducible and the very general fiber has Mumford–Tate group $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$ up to isogeny, for some $h \mid n$. Equivalently, the fiber is isogenous to a product of h-dimensional abelian varieties, each *Hodge-generic* (with no unexpected Hodge classes). Moreover, $h = n$ for a general fibered member of each deformation type with $b_2 \geq 7$. The research objectives are:

RO 1. Determine which h occur, and whether $h < n$ forces $K3^{[n]}$ type, using the Donagi–Markman cubic condition to split the period map along algebraic foliations of the base.

RO 2. Study non-compact Lagrangian fibrations, chiefly the SL_r Hitchin fibration (whose monodromy the PI's methods show to be irreducible), and determine the Mumford–Tate group of its very general fiber via Lefschetz theory for spectral curves.

RO 3. Work toward the hyper-Kähler Bogomolov–Mumford conjecture (a uniruled divisor in every ample linear system), using typical intersections in the moduli of abelian varieties to find families of fibers with an elliptic factor, whose degenerations give rational curves.

Maulik, a leading expert on abelian fibrations, hyper-Kähler varieties, and Hitchin systems, will meet with the PI weekly; at MIT the PI will learn the sheaf-theoretic tools the project needs, such as perverse filtrations, Fourier–Mukai transforms, and moduli of sheaves on K3 surfaces.

Intellectual Merit

Since the SYZ theorem makes Lagrangian fibrations the gateway to classifying hyper-Kähler manifolds, constraints on their fibers bear directly on classification. RO 1 would show that decomposable Lagrangian fibrations are exactly those built from elliptic fibrations, as for Hilbert schemes of points on elliptic K3 surfaces, and would characterize the $K3^{[n]}$ deformation type Hodge-theoretically. RO 2 would give a new Hodge-theoretic invariant of the Hitchin system, a fundamental integrable system in non-abelian Hodge theory and geometric Langlands, complementing recent work on its topology and algebraic cycles. RO 3 would bring functional transcendence to bear on rational curves on hyper-Kähler manifolds, which are tied to their algebraic cycles.

Broader Impacts

The PI will organize a learning seminar on Lagrangian fibrations and Hitchin systems for Boston-area graduate students and postdocs, and seek support for an early-career workshop. Both will be open to all and aimed at researchers without local access to the area, with notes and recorded talks posted publicly. The PI will also volunteer with Boston-area math circles and competitions such as HMMT, favoring programs serving schools with limited access to advanced mathematics, and seek to mentor undergraduates through the MIT Summer Research Program. MIT's teacher-training program for new postdocs will further develop the PI as an educator. Finally, Lagrangian fibrations and Hitchin systems are integrable systems that appear throughout mirror symmetry and mathematical physics, so the results will be useful beyond algebraic geometry.

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The Hodge theory of Lagrangian fibrations, with applications to hyper-Kähler geometry

Applicant: Edward A. Varvak, University of Chicago
Sponsoring scientist: Davesh Maulik, Massachusetts Institute of Technology

1. Introduction and background

Compact Kähler manifolds with trivial canonical bundle, the higher-dimensional analogues of elliptic curves and K3 surfaces, are central objects in algebraic geometry and mathematical physics. By the Beauville–Bogomolov decomposition theorem [Bea83, Bog74], every such manifold is, up to a finite étale cover, a product of complex tori, Calabi–Yau manifolds, and *hyper-Kähler manifolds*: simply connected compact Kähler manifolds X whose holomorphic 2-forms are spanned by a single symplectic form σ. The decomposition extends to varieties with mild (klt) singularities [BGL22], with hyper-Kähler manifolds replaced by their singular analogues, irreducible symplectic varieties.

Hyper-Kähler manifolds are governed by Hodge theory to a remarkable degree: by the global Torelli theorem [Ver13], a deformation class is described by an explicit arithmetic quotient of a period domain, so a single example determines the whole class. The fundamental open problem is instead to determine which deformation classes exist. In each dimension $2n$ only two families are known, $K3^{[n]}$ and Kum_n, along with two sporadic examples, $OG6$ and $OG10$, in dimensions 6 and 10. The guiding conjecture in this direction, the hyper-Kähler SYZ conjecture, was recently proven in a manuscript released by OpenAI [Ope26b], and independently by Engel–Mauri [EM26].

Theorem (Hyper-Kähler SYZ)**.** *Let L be a nontrivial nef line bundle on a hyper-Kähler manifold X which is isotropic for the Beauville–Bogomolov–Fujiki form q_X. Then L is semiample, and a power of L defines a* Lagrangian fibration*: a surjective morphism $f\colon X \to B$ with connected fibers of dimension $n = \frac{1}{2}\dim X$ on which σ vanishes.*

When $b_2(X) \geq 5$, as for all known examples, q_X has isotropic integral classes, so every such deformation class contains a Lagrangian fibration. Combined with the boundedness results of Engel–Filipazzi–Greer–Mauri–Svaldi [EFG$^+$25], it follows that there are only finitely many deformation classes with $b_2 \geq 5$ in each dimension [Ope26b, EM26]. Lagrangian fibrations are therefore the natural gateway to classification. They are also highly constrained: if $f : X \to B$ is a Lagrangian fibration of a hyper-Kähler manifold of dimension $2n$, then its smooth fibers are n-dimensional abelian varieties [Mat99]; the base B is a $\mathbb{Q}$-factorial klt Fano variety of Picard rank one with the intersection cohomology of $\mathbb{P}^n$ [Mat99, SY22], and $B \cong \mathbb{P}^n$ whenever B is smooth [Hwa08, GL14, BS26] or, by a second recent OpenAI manuscript, whenever X is projective [Ope26a], proving a conjecture of Matsushita in the projective case; and f is either isotrivial or of *maximal variation* [Bak25], meaning that the period map $\mathcal{P}\colon B^\circ \to \mathcal{A}_n$, which sends a point of the smooth locus $B^\circ \subset B$ to the isomorphism class of the fiber X_b in the moduli space $\mathcal{A}_n$ of polarized abelian varieties, has n-dimensional image.

The goal of this proposal is to determine where the period map lands, i.e. which abelian varieties, and which families of them, occur as fibers of Lagrangian fibrations. We also hope to use this Hodge-theoretic information to attack structural questions: the classification of Lagrangian fibrations whose fibers decompose (Questions 1 and 2), the extension of the theory to non-compact settings such as Hitchin systems (Questions 3 and 4), and the construction of rational curves and uniruled divisors on hyper-Kähler manifolds (Section 3.3).

2. Past results

My thesis work, supervised by Benjamin Bakker, develops the Hodge theory of Lagrangian fibrations. Let X be a *primitive symplectic variety* of dimension $2n$: a compact symplectic variety with $H^1(X, \mathcal{O}_X) = 0$ and a unique symplectic form up to scaling, such as a hyper-Kähler manifold. Let $f : X \to B$ be a Lagrangian fibration, whose smooth fibers are abelian varieties [Sch20]. Let $f^\circ : X^\circ \to B^\circ$ be the smooth part of f and $V = R^1 f^\circ_* \mathbb{Z}$ the associated variation of Hodge structures, viewed as a representation of $\pi_1(B^\circ)$.

Theorem A ([Var26a]). *If f is not isotrivial, then $V_\mathbb{C}$ is irreducible. If f is isotrivial, then the smooth fibers are isogenous to E^n for an elliptic curve E (first proven in [KLM26]), and $V_K = U_1 \oplus U_2$ splits into two geometrically irreducible summands, where K is the endomorphism field of E.*

Theorem B ([Var26b]). *If f is not isotrivial, then for very general $b \in B$ the Mumford–Tate group of X_b is isogenous over $\mathbb{Q}$ to $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$ for some $h \mid n$. Equivalently, X_b is isogenous to a product $A_1 \times \cdots \times A_{n/h}$ of pairwise non-isogenous h-dimensional abelian varieties, each* Hodge-generic*, i.e. carrying no Hodge classes beyond those forced by its polarization.*

Theorem C ([Var26b]). *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds with $b_2 \geq 7$ which contains a Lagrangian fibration. If $X \in \mathcal{X}$ is general among members admitting a Lagrangian fibration $f\colon X \to B$, then the very general fiber of f is Hodge-generic: $\mathbf{MT}(X_b) = \mathrm{GSp}_{2n}$.*

These results sharpen Bakker's theorem: the period image is not only n-dimensional but Hodge-generic either in $\mathcal{A}_n$ or in a product locus $\mathcal{A}_h^{n/h}$, and in the general case it is the former. The proof of Theorem B hinges on the interplay between the connected monodromy group, which preserves each factor A_i, and the finite part of the monodromy, which by Theorem A must permute the factors transitively. The hypothesis $b_2 \geq 7$ holds for every known deformation type, and by the SYZ theorem every such deformation class contains a Lagrangian fibration.

Personal accomplishments

I've been invited to speak about my research at the Institute for Advanced Study's Research Seminar (November 2025), GTA Philadelphia (May 2026), and the upcoming Special Session on Higgs Bundles at JMM 2027. I've received the Kuga–Sah Award for Excellence in Mathematics from the Stony Brook University Department of Mathematics (2022). I have received the UIC RTG grant for six semesters throughout my time at UIC. I received the Bernard Kurtin Award for research in mathematics from the UIC Department of Mathematics, Statistics, and Computer Science (2025).

3. Research objectives, methods, and significance

We will break our research plan into three primary objectives, which we order by the expected difficulty of the task in question.

RO 1. Determine which decompositions of the Mumford–Tate group occur for Lagrangian fibers.
RO 2. Understand the Hodge theory of non-compact Lagrangian fibrations.
RO 3. Work towards the Bogomolov–Mumford conjecture for hyper-Kähler manifolds.

In this section, we will go through each of them, laying out a motivation and an approach.

3.1. Which decompositions of the Mumford–Tate group occur?

Theorem B leaves open which values of h actually occur. If $S \to \mathbb{P}^1$ is a non-isotrivial elliptic K3 surface, the induced map $S^{[n]} \to (\mathbb{P}^1)^{(n)} \cong \mathbb{P}^n$ is a Lagrangian fibration whose fibers are isogenous to products of n elliptic curves, so $h = 1$. If singularities are allowed, every h occurs: given a Lagrangian fibration $Y \to \mathbb{P}^h$ with Hodge-generic fibers (Theorem C), the symmetric power $Y^{(k)} \to (\mathbb{P}^h)^{(k)}$

is a Lagrangian fibration of a primitive symplectic variety with $\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k$. For $h \geq 2$, however, the base $(\mathbb{P}^h)^{(k)}$ does not have the intersection cohomology of projective space (e.g. $\dim \mathrm{IH}^4((\mathbb{P}^2)^{(2)}) = 2$), so it cannot be the base of a Lagrangian fibration of a hyper-Kähler manifold.

Question 1. *Is there a hyper-Kähler manifold (or irreducible symplectic variety) with a Lagrangian fibration such that* $\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$ *with* $1 < h < n$*?*

Question 2. *If X is hyper-Kähler and the very general fiber is not Hodge-generic, i.e. $h < n$, is X of $K3^{[n]}$-type?*

I expect a negative answer to Question 1: whenever the fibers are not Hodge-generic, the fibration should be governed by families of elliptic curves, as it is for $S^{[n]}$. A complete answer would show that decomposable Lagrangian fibrations are exactly those built from elliptic fibrations and give a Hodge-theoretic characterization of the $K3^{[n]}$ deformation type.

Approach. Suppose $h < n$, set $k = n/h$, and pass to the finite étale cover $B' \to B^\circ$ over which V splits as $V_1 \oplus \cdots \oplus V_k$. The symplectic form identifies T_bB with $H^{1,0}(X_b)$, and by Donagi–Markman [DM96] the differential of the period map is a symmetric cubic form on T_bB. Because the period map respects the splitting, this cubic form is a sum of cubic forms on the summands $E_i \subset T_bB$ corresponding to the $H^{1,0}(V_{i,b})$. Consequently each partial period map $\mathcal{P}_i \colon B' \to \mathcal{A}_h$ depends only on the E_i-directions, the E_i integrate to algebraic foliations, and $(\mathcal{P}_1, \ldots, \mathcal{P}_k)$ maps B' dominantly and generically finitely onto a product $Z_1 \times \cdots \times Z_k$ of h-dimensional varieties. This is exactly the structure of $(\mathbb{P}^1)^n \to \mathbb{P}^n$ in the example above. I plan to extend this structure across the discriminant locus and combine it with the constraints on B (Fano, Picard rank one, $\mathrm{IH}^*(B) \cong H^*(\mathbb{P}^n)$, and now $B \cong \mathbb{P}^n$ when X is a projective hyper-Kähler manifold [Ope26a]) and an analysis of the branching of $B' \to B$ to rule out $1 < h < n$. When $h = 1$, I will use the factors to construct an elliptic K3 surface S and compare X with a moduli space of sheaves on S, identifying its deformation type. Recent work of Saccà on compactifications of Lagrangian fibrations [Sac25] will be an important input.

3.2. Lagrangian fibrations beyond the compact case

Many important Lagrangian fibrations are not compact: integrable systems from mathematical physics, local models near singular fibers, and above all the Hitchin fibration $h \colon \mathcal{M}_{\mathrm{Dol}} \to \mathbb{A}$ on the moduli space of Higgs bundles on a curve C [Hit87], which is central to non-abelian Hodge theory and the geometric Langlands program. My compact methods use Bakker's dichotomy, which fails here: h is equivariant for the scaling action $(E, \theta) \mapsto (E, \lambda\theta)$, so its period map is constant along $\mathbb{C}^*$-orbits and never has maximal variation.

Question 3. *Let $f \colon M \to B$ be a proper Lagrangian fibration of a (not necessarily compact) holomorphic symplectic manifold, of maximal variation. Is the Mumford–Tate group of its very general fiber isogenous to* $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h_1} \times \cdots \times \mathrm{Sp}_{2h_k})$ *with* $h_1 + \cdots + h_k = n$*?*

I expect the foliation and monodromy techniques above to extend to this setting, possibly under mild hypotheses on the singular fibers. For the Hitchin fibration of SL_r-Higgs bundles, whose regular fibers are Prym varieties of spectral covers $\Sigma_a \to C$, I can adapt the methods of [Var26a] to show that $R^1h_*\mathbb{C}$ is irreducible over the regular locus $\mathbb{A}^\circ$, despite the failure of maximal variation.

Question 4. *Is the very general* SL_r*-Hitchin fiber Hodge-generic? If not, which products of symplectic groups occur as its Mumford–Tate group?*

Mumford–Tate groups govern the Hodge classes, and conjecturally the algebraic cycles, on the fibers. This complements recent work of Maulik–Shen–Yin on algebraic cycles and Fourier transforms for Hitchin systems [MSY24, MSY23], and gives a new Hodge-theoretic invariant of a fundamental

integrable system.

Approach. Spectral curves form an open subset of a linear system on the compactified cotangent bundle $\mathbb{P}(K_C \oplus \mathcal{O}_C)$, so Lefschetz theory is available: the monodromy is generated by Picard–Lefschetz transformations in vanishing cycles, and Zariski density results [Del74, Lemma 4.4.2[a]] apply. Since the connected monodromy group is a normal subgroup of the derived Mumford–Tate group [And92], showing that the monodromy on the Prym part is Zariski-dense in Sp_{2d}, $d = \dim \mathbb{A}$, would prove Hodge-genericity. The difficulties are that the Hitchin base is only part of a linear system and that the relevant vanishing cohomology is that of the Prym; here we will try to apply the work of Maulik and his collaborators on the topology of Hitchin fibrations [dCMS22, MS24].

3.3. Rational curves and the Bogomolov–Mumford conjecture

By a theorem of Bogomolov and Mumford [MM83], every ample linear system on a K3 surface contains a divisor supported on rational curves. The analogue for hyper-Kähler manifolds is closely tied to the study of algebraic cycles on hyper-Kähler varieties [CMP19]; it is known for the deformation types $K3^{[n]}$ and Kum_n but open in general.

Conjecture (Hyper-Kähler Bogomolov–Mumford)**.** *If X is a projective hyper-Kähler manifold and L is an ample line bundle on X, then $|L|$ contains a uniruled divisor.*

Approach. Let X be general among Lagrangian-fibered members of its deformation class, which exist by the SYZ theorem, so that by Theorem C the period image $\mathcal{P}(B^\circ) \subset \mathcal{A}_n$ is n-dimensional and Hodge-generic. Abelian varieties isogenous to a product $E \times A'$ form the Hecke translates of $\mathcal{A}_1 \times \mathcal{A}_{n-1}$, which has codimension $n-1$, so their expected intersection with the period image is one-dimensional. Results on the existence and density of such typical intersections [KU23, ES25], complementing the Ax–Schanuel theorem [BT19], which controls the atypical ones, should produce a dense set of curves $C^\circ \subset B^\circ$ over which the fibers are isogenous to $E_t \times A'_t$. Over the boundary points of such a curve I expect the elliptic factor to degenerate to a nodal rational curve while A'_t remains an abelian variety, producing uniruled subvarieties in the singular fibers of f. We need to show that these rational curves deform to a general polarized deformation of X, which no longer carries a Lagrangian fibration, using the deformation theory of rational curves on hyper-Kähler manifolds [CMP19]. Then we hope to follow the K3 picture, where an ample class is a combination $a\sigma + bF$ of a section and a fiber. We will try to decompose a divisor in $|L|$ into uniruled horizontal and vertical parts that deform with X.

4. Career development and choice of sponsoring scientist and host institution

My goal is a faculty career at a research university, leading a research program on the Hodge theory of Lagrangian fibrations and its interactions with moduli theory, enumerative geometry, and non-abelian Hodge theory. My training so far is centered on Hodge theory and period maps. I have chosen to pursue the Research Fellowship because full-time research support suits the long-term, higher-risk parts of this program (such as Section 3.3) and the substantial new background they may require. The fellowship will allow me to:

- learn the sheaf-theoretic and enumerative tools that the proposed questions require: perverse filtrations, Fourier transforms on abelian fibrations, moduli of sheaves on K3 surfaces, and Gromov–Witten theory;
- establish my independence, pursuing the more challenging Question 4 and Section 3.3, starting collaborations at MIT and in the Boston area, and traveling to conferences and research visits supported by the fellowship allowance;

- develop the non-research skills of a faculty position by helping organize MIT's algebraic geometry and preprint seminars, taking part in the department's intensive teacher-training program for new postdocs and its workshops on grant writing and undergraduate advising, and mentoring students through the programs described in Section 5.

My sponsoring scientist, Davesh Maulik, is a leading expert on abelian fibrations, hyper-Kähler varieties, and Hitchin systems, which serve as the backbone of this proposal. A central theme of his recent work is that Lagrangian fibrations form a distinguished class among abelian fibrations, whose cohomology behaves most like that of abelian schemes [MSY23, MSY26]; Theorems A and B, whose analogues fail for general abelian fibrations, are Hodge-theoretic results of the same kind. His work bears directly on each objective. With Shen and Yin he developed the interaction of Fourier–Mukai transforms with perverse filtrations for abelian fibrations [MSY23], and with Shen, Yin, and Zhang he proved the D-equivalence conjecture for hyper-Kähler varieties of $K3^{[n]}$-type [MSYZ24]. These tools, together with his expertise on moduli of sheaves on K3 surfaces, will help to recognize the hyper-Kähler manifolds in Question 2 as belonging to the deformation space $K3^{[n]}$.

Professor Maulik is a central figure in the topology of Hitchin fibrations, including the proof of the $P = W$ conjecture [dCMS22, MS24] and the study of algebraic cycles on Hitchin systems [MSY24]. His insight will be invaluable to helping attack Question 4. Furthermore, his work on Noether–Lefschetz theory [MP13, KMPS10] and on curves on K3 surfaces [MPT10] sits at the interface of Hodge loci and rational curves that Section 3.3 explores. He will be in residence throughout the fellowship, and we will meet weekly.

MIT offers a large and active community in algebraic geometry and neighboring fields, and the Boston area has several Hodge theorists, notably Mihnea Popa at Harvard, whose work on Hodge modules complements Maulik's sheaf-theoretic methods. Moving from Chicago, where I trained in Hodge theory with Bakker, to MIT, with its strengths in moduli theory, enumerative geometry, and geometric representation theory, is precisely the broadening my research program needs.

5. Broader Impacts

Building research communities. As a graduate student I co-organized the Graduate Algebraic Geometry Seminar for two semesters, organized a reading group on the Weil conjectures, and organized a seminar on Lie groups. At MIT I will organize a learning seminar on Lagrangian fibrations and Hitchin systems, open to graduate students and postdocs across the Boston area, and seek support for a small workshop for early-career researchers. Both will be open to everyone and designed to reach researchers who lack local access to this area, including those from groups underrepresented in mathematics. Seminar notes and recorded talks will be posted publicly.

Teaching and outreach. I am currently teaching Calculus II at the University of Chicago as a graduate student lecturer, after five semesters as a teaching assistant at UIC, a Hispanic-Serving Institution where many students are the first in their families to attend college. I have also judged the QED competition of the Math Circles of Chicago; in Boston I will continue this outreach by volunteering with local math circles and competitions such as HMMT, favoring programs that serve schools with limited access to advanced mathematics. I will also seek to mentor undergraduates through the MIT Summer Research Program, which brings students from institutions across the country to MIT to prepare them for graduate study.

Broader scientific impact. Lagrangian fibrations and Hitchin systems are algebraic integrable systems that appear throughout mirror symmetry and mathematical physics; results on their monodromy and periods will be useful well beyond algebraic geometry.

References

[And92] Yves André. Mumford–Tate groups of mixed Hodge structures and the theorem of the fixed part. *Compositio Mathematica*, 82(1):1–24, 1992.

[Bak25] Benjamin Bakker. A short proof of a conjecture of Matsushita. *Advances in Mathematics*, 482:110554, 2025. doi:10.1016/j.aim.2025.110554.

[Bea83] Arnaud Beauville. Variétés Kähleriennes dont la première classe de Chern est nulle. *Journal of Differential Geometry*, 18(4):755–782, 1983.

[BGL22] Benjamin Bakker, Henri Guenancia, and Christian Lehn. Algebraic approximation and the decomposition theorem for Kähler Calabi–Yau varieties. *Inventiones Mathematicae*, 228:1255–1308, 2022. doi:10.1007/s00222-022-01096-y.

[Bog74] Fedor A. Bogomolov. On the decomposition of Kähler manifolds with trivial canonical class. *Mathematics of the USSR-Sbornik*, 22(4):580–583, 1974.

[BS26] Benjamin Bakker and Christian Schnell. A Hodge-theoretic proof of Hwang's theorem on base manifolds of Lagrangian fibrations. *Annales de l'Institut Fourier*, 2026. Online first. doi:10.5802/aif.3786.

[BT19] Benjamin Bakker and Jacob Tsimerman. The Ax–Schanuel conjecture for variations of Hodge structures. *Inventiones Mathematicae*, 217(1):77–94, 2019.

[CMP19] François Charles, Giovanni Mongardi, and Gianluca Pacienza. Families of rational curves on holomorphic symplectic varieties and applications to zero-cycles, 2019. arXiv:1907.10970.

[dCMS22] Mark Andrea A. de Cataldo, Davesh Maulik, and Junliang Shen. Hitchin fibrations, abelian surfaces, and the $P = W$ conjecture. *Journal of the American Mathematical Society*, 35, 2022. arXiv:1909.11885.

[Del74] Pierre Deligne. La conjecture de Weil. I. *Publications Mathématiques de l'IHÉS*, 43:273–307, 1974.

[DM96] Ron Donagi and Eyal Markman. Spectral covers, algebraically completely integrable, Hamiltonian systems, and moduli of bundles. In *Integrable systems and quantum groups (Montecatini Terme, 1993)*, volume 1620 of *Lecture Notes in Mathematics*, pages 1–119. Springer, Berlin, 1996.

[EFG+25] Philip Engel, Stefano Filipazzi, François Greer, Mirko Mauri, and Roberto Svaldi. Boundedness of some fibered K-trivial varieties, 2025. URL: https://arxiv.org/abs/2507.00973, arXiv:2507.00973.

[EM26] Philip Engel and Mirko Mauri. Hyperkähler SYZ conjecture, 2026. arXiv:2610.12277.

[ES25] Sebastian Eterović and Thomas Scanlon. Likely intersections. *Forum of Mathematics, Sigma*, 13:e199, 2025. doi:10.1017/fms.2025.10114.

[GL14] Daniel Greb and Christian Lehn. Base manifolds for Lagrangian fibrations on hyperkähler manifolds. *International Mathematics Research Notices*, 2014(19):5483–5487, 2014. doi:10.1093/imrn/rnt133.

[Hit87] Nigel Hitchin. Stable bundles and integrable systems. *Duke Mathematical Journal*, 54(1):91–114, 1987.

[Hwa08] Jun-Muk Hwang. Base manifolds for fibrations of projective irreducible symplectic manifolds. *Inventiones mathematicae*, 174(3):625–644, 2008. doi:10.1007/s00222-008-0143-9.

[KLM26] Yoon-Joo Kim, Radu Laza, and Olivier Martin. Isotrivial Lagrangian fibrations of compact hyper-Kähler manifolds. *Journal de Mathématiques Pures et Appliquées*, 205:103810, 2026. doi:10.1016/j.matpur.2025.103810.

[KMPS10] Albrecht Klemm, Davesh Maulik, Rahul Pandharipande, and Emanuel Scheidegger. Noether–Lefschetz theory and the Yau–Zaslow conjecture. *Journal of the American Mathematical Society*, 23(4):1013–1040, 2010.

[KU23] Nazim Khelifa and David Urbanik. Existence and density of typical Hodge loci, 2023. arXiv:2303.16179.

[Mat99] Daisuke Matsushita. On fibre space structures of a projective irreducible symplectic manifold. *Topology*, 38(1):79–83, 1999. doi:10.1016/S0040-9383(98)00003-2.

[MM83] Shigefumi Mori and Shigeru Mukai. The uniruledness of the moduli space of curves of genus 11. In *Algebraic geometry (Tokyo/Kyoto, 1982)*, volume 1016 of *Lecture Notes in Mathematics*, pages 334–353. Springer, Berlin, 1983.

[MP13] Davesh Maulik and Rahul Pandharipande. Gromov–Witten theory and Noether–Lefschetz theory. In *A celebration of algebraic geometry*, volume 18 of *Clay Mathematics Proceedings*, pages 469–507. American Mathematical Society, 2013.

[MPT10] Davesh Maulik, Rahul Pandharipande, and Richard P. Thomas. Curves on K3 surfaces and modular forms. *Journal of Topology*, 3(4):937–996, 2010.

[MS24] Davesh Maulik and Junliang Shen. The $P = W$ conjecture for GL_n. *Annals of Mathematics*, 200, 2024. arXiv:2209.02568.

[MSY23] Davesh Maulik, Junliang Shen, and Qizheng Yin. Perverse filtrations and Fourier transforms, 2023. Acta Math. arXiv:2308.13160.

[MSY24] Davesh Maulik, Junliang Shen, and Qizheng Yin. Algebraic cycles and Hitchin systems, 2024. To appear in Ann. Sci. Éc. Norm. Supér. arXiv:2407.05177.

[MSY26] Davesh Maulik, Junliang Shen, and Qizheng Yin. Dualizable abelian fibrations, 2026. arXiv:2602.19318.

[MSYZ24] Davesh Maulik, Junliang Shen, Qizheng Yin, and Ruxuan Zhang. The D-equivalence conjecture for hyper-Kähler varieties via hyperholomorphic bundles, 2024. To appear in Invent. Math. arXiv:2408.14775.

[Ope26a] OpenAI. Projective-space bases of Lagrangian fibrations. OpenAI Math Release preprint, 2026. Released October 6, 2026. URL: https://github.com/openai/math/blob/main/preprints/Projective-Space-Bases-of-Lagrangian-Fibrations-September-23-2026/main.pdf.

[Ope26b] OpenAI. The strong hyperkähler SYZ conjecture. OpenAI Math Release preprint, 2026. Released October 6, 2026. URL: https://github.com/openai/math/blob/main/preprints/The-Strong-Hyperkahler-SYZ-Conjecture-September-23-2026/main.pdf.

[Sac25] Giulia Saccà. Compactifying Lagrangian fibrations, 2025. arXiv:2411.06505.

[Sch20] Martin Schwald. Fujiki relations and fibrations of irreducible symplectic varieties. *Épijournal de Géométrie Algébrique*, 4(7), 2020. doi:10.46298/epiga.2020.volume4.4557.

[SY22] Junliang Shen and Qizheng Yin. Topology of Lagrangian fibrations and Hodge theory of hyper-Kähler manifolds. *Duke Mathematical Journal*, 2022. arXiv:1812.10673.

[Var26a] Edward Varvak. The monodromy of compact Lagrangian fibrations. *International Mathematics Research Notices*, 2026(7):rnag057, 03 2026. doi:10.1093/imrn/rnag057.

[Var26b] Edward Varvak. The Mumford–Tate group of a very general Lagrangian fiber, 2026. arXiv:2610.02481.

[Ver13] Misha Verbitsky. Mapping class group and a global Torelli theorem for hyperkähler manifolds. *Duke Mathematical Journal*, 162(15):2929–2986, 2013.

SUMMARY PROPOSAL BUDGET

YEAR 1

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
AWARD NO.		

ORGANIZATION: Varvak, Edward A

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Edward Varvak

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Edward Varvak - Principal Inv	0.0			0	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	0.0			0	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (0) GRADUATE STUDENTS				0	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				0	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				0	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				0	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				0	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 30,000					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 35,000					
TOTAL NUMBER OF PARTICIPANTS (1) TOTAL PARTICIPANT COSTS				65,000	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				0	
5. SUBAWARDS				0	
6. OTHER				0	
TOTAL OTHER DIRECT COSTS				0	
H. TOTAL DIRECT COSTS (A THROUGH G)				65,000	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				0	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				65,000	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				65,000	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Edward Varvak

ORG. REP. NAME*

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
AWARD NO.		

ORGANIZATION
Varvak, Edward A

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR
Edward Varvak

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Edward Varvak - Principal Inv	0.0			0	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	0.0			0	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (0) GRADUATE STUDENTS				0	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				0	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				0	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				0	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				0	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 20,000					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 35,000					
TOTAL NUMBER OF PARTICIPANTS (1) TOTAL PARTICIPANT COSTS				55,000	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				0	
5. SUBAWARDS				0	
6. OTHER				0	
TOTAL OTHER DIRECT COSTS				0	
H. TOTAL DIRECT COSTS (A THROUGH G)				55,000	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				0	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				55,000	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				55,000	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME
Edward Varvak

ORG. REP. NAME*

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

***ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET**

SUMMARY PROPOSAL BUDGET

YEAR 3

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
AWARD NO.		

ORGANIZATION: Varvak, Edward A

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Edward Varvak

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Edward Varvak - Principal Inv	0.0			0	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	0.0			0	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (0) GRADUATE STUDENTS				0	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				0	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				0	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				0	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				0	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 35,000					
TOTAL NUMBER OF PARTICIPANTS (1) TOTAL PARTICIPANT COSTS				35,000	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				0	
5. SUBAWARDS				0	
6. OTHER				0	
TOTAL OTHER DIRECT COSTS				0	
H. TOTAL DIRECT COSTS (A THROUGH G)				35,000	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				0	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				35,000	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				35,000	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Edward Varvak

ORG. REP. NAME*

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 4

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
AWARD NO.		

ORGANIZATION: Varvak, Edward A

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Edward Varvak

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Edward Varvak - Principal Inv	0.0			0	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	0.0			0	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (0) GRADUATE STUDENTS				0	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				0	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				0	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				0	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				0	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 35,000					
TOTAL NUMBER OF PARTICIPANTS (1) TOTAL PARTICIPANT COSTS				35,000	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				0	
5. SUBAWARDS				0	
6. OTHER				0	
TOTAL OTHER DIRECT COSTS				0	
H. TOTAL DIRECT COSTS (A THROUGH G)				35,000	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				0	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				35,000	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				35,000	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Edward Varvak

ORG. REP. NAME*

FOR NSF USE ONLY		
INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
AWARD NO.		

ORGANIZATION: Varvak, Edward A

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: Edward Varvak

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. Edward Varvak - Principal Inv	0.0			0	
2.					
3.					
4.					
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)					
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	0.0			0	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	0.0			0	
3. (0) GRADUATE STUDENTS				0	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				0	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				0	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				0	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				0	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				0	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 50,000					
2. TRAVEL 0					
3. SUBSISTENCE 0					
4. OTHER 140,000					
TOTAL NUMBER OF PARTICIPANTS (4) TOTAL PARTICIPANT COSTS				190,000	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				0	
3. CONSULTANT SERVICES				0	
4. COMPUTER SERVICES				0	
5. SUBAWARDS				0	
6. OTHER				0	
TOTAL OTHER DIRECT COSTS				0	
H. TOTAL DIRECT COSTS (A THROUGH G)				190,000	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				0	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				190,000	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				190,000	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: Edward Varvak

ORG. REP. NAME*

FOR NSF USE ONLY		
INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

Budget Justification

Budget justification is not applicable to this proposal.

Budget Justification

Budget justification is not applicable to this proposal.

IDENTIFYING INFORMATION:

NAME: Varvak, Edward Ariel

ORCID iD: https://orcid.org/0009-0004-8413-5873

POSITION TITLE: Graduate Student

PRIMARY ORGANIZATION AND LOCATION: University of Chicago, Chicago, Illinois, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Chicago, Chicago, Illinois, United States	PHD	10/2026 - 06/2027	Mathematics
University of Illinois Chicago, Chicago, Illinois, United States	PHD	08/2022 - 09/2026	Mathematics
Stony Brook University, Stony Brook, New York, United States	BS	08/2018 - 05/2022	Mathematics

Appointments and Positions

2026 - present Graduate Student, University of Chicago, Chicago, Illinois, United States

2022 - 2026 Graduate Student, University of Illinois Chicago, Chicago, Illinois, United States

2022 - 2026 Fellow, University of Illinois Chicago, Chicago, Illinois, United States

Products

Products Most Closely Related to the Proposed Project

1. Varvak E. The Mumford--Tate group of a very general Lagrangian fiber. Version: v1. arXiv [Preprint]. 2026 October 01: [23 pages]. Available from: https://arxiv.org/abs/2610.02481 DOI: 10.48550/arXiv.2610.02481
2. Varvak E. The monodromy of compact Lagrangian fibrations. The monodromy of compact Lagrangian fibrations. 2026 April 01; 2026(7). DOI: 10.1093/imrn/rnag057

Other Significant Products, Whether or Not Related to the Proposed Project

1. Varvak E. Recovering nodal plane curves from their bitangent lines. Github [Preprint]. 2022 May 01: [10 pages]. Available from: https://evarvak.github.io/documents/SBU_Honors_Thesis.pdf

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Varvak, Edward Ariel in SciENcv on 2026-10-20 19:12:31

Data Not Available

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Varvak, Edward Ariel

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0009-0004-8413-5873

*POSITION TITLE: Graduate Student

*ORGANIZATION AND LOCATION: University of Chicago, Chicago, Illinois, United States

Proposals/Active Projects

***Proposal/Active Project Title:**	MSPRF: The Hodge theory of Lagrangian fibrations, with applications to hyper-Kahler geometry (this proposal)
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	Massachusetts Institute of Technology, Cambridge, MA, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2027
***Proposal/Active Project End Date: (MM/YYYY):**	08/2031
***Total Anticipated Proposal/Project Amount:**	$190,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	8
2028	8
2029	8

***Overall Objectives:**

The project studies the Hodge theory of Lagrangian fibrations of hyper-Kahler manifolds and related holomorphic symplectic varieties. Its objectives are
(1) to determine which decompositions of the Mumford--Tate group occur for fibers of Lagrangian fibrations, and whether a decomposable fiber forces a hyper-Kahler manifold to be deformation equivalent to a Hilbert scheme of points on a K3 surface;
(2) to extend this Hodge theory to non-compact Lagrangian fibrations, chiefly the Hitchin fibration, by determining the Mumford--Tate group of its very general fiber; and
(3) to work toward the hyper-Kahler Bogomolov--Mumford conjecture by producing rational curves from degenerations of fibers with an elliptic factor.

***Statement of Potential Overlap:**

There is no potential overlap.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Varvak, Edward in SciENcv on 2026-10-20 19:26:32

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Varvak, Edward A	University of Chicago, Chicago, IL	
		University of Illinois Chicago, Chicago, IL	09/27/26
		MIT, Cambridge, MA (proposed host)	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Varvak, Anna	Aunt		
R	Geraschenko, Anton	Uncle		

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Bakker, Benjamin	University of Chicago, Chicago, IL	bakker.uchicago@gmail.com

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date

Synergistic Activities

Edward A. Varvak

1. **Seminar organization.** Co-organizer of the Graduate Algebraic Geometry Seminar at the University of Illinois Chicago (UIC), Fall 2025–Spring 2026, a student-run seminar in which graduate students give talks on algebraic geometry.

2. **Learning seminars.** Organizer of two learning seminars for graduate students at UIC: a reading group on the Weil conjectures (Fall 2025) and a seminar on Lie groups (Fall 2024). Speaker at the UIC Graduate Student Colloquium (February 2025), presenting research to a general audience of graduate students.

3. **Mathematics outreach.** Judge at QED, Chicago's Youth Math Symposium, run by the Math Circles of Chicago (December 2025), at which students in grades 5–12 present original projects in mathematics and computer science.

4. **Teaching.** Graduate student lecturer for Calculus II at the University of Chicago (Fall 2026). Teaching assistant for five semesters at UIC, a Hispanic-Serving Institution where many students are the first in their families to attend college, in Finite Mathematics for Business (four semesters) and Calculus II (one semester).

5. **Mentoring graduate students.** Member of the geometry advisor panel of the UIC Department of Mathematics, Statistics, and Computer Science, and an informal mentor to younger graduate students navigating graduate school.

Data Management and Sharing Plan

Submitted to:
Directorate for Mathematical and Physical Sciences (MPS), MPS - Core Programs (MCP), Workforce (MSPRF) MathSciPDFellow

1. Research articles and preprints

Data or Research Product Category General Information

Data or Research Product Category: **Research articles and preprints**

Description: **Research papers presenting the results of the project (theorems, proofs, and examples), and an expository survey if one is written.**

Access Policies and Limitations

Not applicable

Data Standards and Metadata

Data/Metadata Standard: **PDF**

Additional Details: **LaTeX source will be deposited on arXiv and on github.**

Data or Research Product Provenance

Data Source: **Original research by the PI**

Public Archiving

Data Sharing Location: **arXiv and NSF-PAR**

Additional Details: **All preprints will be stored on arXiv, and all accepted manuscripts will be submitted to the NSF Public Access Repository (NSF-PAR)**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Additional Details: **Each preprint will be posted to arXiv no later than its submission to a journal. Once a manuscript is accepted, it will be uploaded to NSF-PAR.**

Data Availability

Data Availability Period: **Indefinitely**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Edward Varvak**

Role: **Proposed Postdoctoral Fellow**

Email Address: **varvak@uchicago.edu**

Data Not Available

Data Not Available

Data Not Available

Data Not Available

Data Not Available

Data Not Available

MSPRF Sponsoring Scientist Statement

Proposed Sponsoring Scientific Advisor:
Name: Davesh Maulik
Title: Professor
E-mail address: maulik@mit.edu
Telephone: 617- 253-1796
Proposed Fellowship Institution:
Organization name: MIT
Department: Mathematics
Street: 77 Massachusetts Avenue
City: Cambridge
State: MA
ZIP: 02139
Country: USA

This note is a sponsoring scientist statement for Edward Varvak to receive an NSF postdoctoral fellowship at MIT, starting 2026-2027.

Let me begin by discussing the compatibility of myself as sponsoring scientist with Varvak's research focus and proposal. Varvak's interests involve hyperkahler geometry and Lagrangian fibrations. He has written a series of papers studying Hodge-theoretic constraints on the fibers of a compact Lagrangian fibration; specifically, there are strong conditions of the Mumford-Tate group of the generic fiber of such families. He has a number of plans to study these constraints further and examine the noncompact setting. These questions overlap with my own recent research work, in which me and my collaborators are interested in studying special properties of the topology of Lagrangian fibrations (when compared to general abelian fibrations). As a result, I expect there will be many points of common research contact.

Furthermore, MIT (and the Boston area overall) is a natural fit for these interests. There are several geometers with Hodge-theoretic interest in the greater Boston area (notably Mihnea Popa)

In terms of the logistics of sponsorship, I expect to be regularly available during the fellowship period. In particular, I have no long-term travel plans during this period. If he receives the fellowship, I would meet him on a regular basis to discuss his research progress and any related questions that arise. If he's interested, he could help organize algebraic geometry seminar and the preprint seminar. Beyond the research aspects of his training, MIT offers ample opportunities for non-research professional development; most notably, the department offers an intensive teacher training program for new postdocs, which he would be encouraged to participate in, as well as workshops in grant-writing and undergraduate advising.

Data Not Available

Data Not Available